MODUL VII

Testing, Debugging dan Security

Tujuan

- Mahasiswa dapat menulis dan menjalankan Feature Test untuk endpoint CRUD.
- Mahasiswa dapat menggunakan Log facade untuk mencatat operasi CRUD.
- Mahasiswa dapat mengkonfigurasi rate limiting pada route API.
- Mahasiswa dapat menjalankan `composer audit` dan mengidentifikasi/vulnerabilitas.
- Mahasiswa memahami dasar debugging dengan dd() dan Log.
- skenario bahwa hanya admin dapat melakukan delete.

Materi

1. Testing Dasar (PHPUnit & Laravel)

1.1. Lokasi tests/Feature dan tests/Unit

1.2. Feature Test: simulasi HTTP request ke endpoint

- Artisan: `php artisan make:test ItemApiTest --unit=false`
- Assertions:

```
this->getJson('/api/v1/items')
    ->assertStatus(200)
    ->assertJsonStructure(['status','data'])

this->postJson(...)
    ->assertStatus(201)
    ->assertJsonFragment([...])

$this->deleteJson(...)->assertStatus(403)
```

1.3. Menjalankan test: php artisan test atau vendor/bin/phpunit

2. Debugging & Logging

2.1. Debugging:

- dd(), dump(), ray() (pakai spatie/laravel-ray)
- Xdebug + IDE integration

2.2. Logging:

- Facade Log (Log::info, Log::error)
- Channel: daily, single, stack (config/logging.php)
- Penulisan log di Service Layer untuk setiap aksi penting

3. Keamanan Lanjutan

3.1. CSRF

- API routes (`routes/api.php`) tidak memerlukan CSRF token

- Web routes (**routes/web.php**) dilindungi oleh **VerifyCsrfToken** middleware

3.2. Rate Limiting

- Middleware **throttle:rate,minutes**
- Default di **app/Http/Kernel.php** pada group 'api': **'throttle:api'** (default 60 req/1 mnt)
- Custom: **Route::middleware('throttle:100,1')->group(…)**

3.3. Audit Dependency

- **composer audit** (scan vulnerability)
- **composer audit --format=json**
- Update package:
  composer update vendor/package --with-dependencies

Praktikum

1. Pastikan Anda switch dari branch main ke branch `feature/testing-debug-security`:

```
cd C:\laragon\www\inventory
git checkout main
git pull origin main
git checkout -b feature/testing-debug-security
```

2. Tambah Logging di Service Layer

2.1. Buka app/Services/ItemService.php, tambahkan
use Illuminate\Support\Facades\Log;

2.2. Contoh untuk method create:

```
public function create(array $data) {
  $item = Item::create($data);
  Log::info('Item created', [
    'id'=>$item->id,
    'data'=>$data
  ]);
  return $item;
}
```

2.3. Lakukan serupa di update() dan delete():

```
Log::info('Item updated', ['id'=>$id,'changes'=>$data]);
Log::info('Item deleted', ['id'=>$id]);
```

3. Konfigurasi Rate Limit

3.1. Buka app/Http/Kernel.php, cari **$middlewareGroups['api']**

3.2. Ubah **'throttle:api'** menjadi **'throttle:60,1'** atau tambahkan middleware custom:

```
protected $middlewareGroups = [
 'api' => ['throttle:60,1',
   \Illuminate\Routing\Middleware\SubstituteBindings::class,
]];
```

3.3. Atau di routes/api.php:

```
Route::prefix('v1')->middleware([
    'auth:sanctum',
    'throttle:60,1'
  ])->group(function(){
  // routes...
});
```

4. Debugging
   - Tambahkan dd($variable) di controller atau service untuk inspect data
   - Jalankan request via Postman/Browser dan perhatikan output
5. Tulis Feature Tests

5.1. Buat test file:

```
php artisan make:test ItemApiTest
```

File: tests/Feature/ItemApiTest.php

```
<?php
namespace Tests\Feature;
use Tests\TestCase;
use App\Models\User;
use App\Models\Category;
use Illuminate\Foundation\Testing\RefreshDatabase;
class ItemApiTest extends TestCase {
  use RefreshDatabase;
  protected $user;
  protected $admin;
  protected $category;
  protected function setUp() {
    parent::setUp();
    Category::factory()->create([
      "id" => 1, "name" => "Electronics"
    ]);
    $this->user = User::factory()->create([
      "role" => "user"
    ]);
    $this->admin = User::factory()->create([
      "role" => "admin"
    ]);
  }
  public function test_guest_cannot_access_items(){
    $this->getJson("/api/v1/items")->assertStatus(401);
  }
  public function test_user_can_list_items(){
    $token = $this->user->createToken("api-token")
        ->plainTextToken;
    $this->withHeader("Authorization", "Bearer $token")
        ->getJson("/api/v1/items")
        ->assertStatus(200)
        ->assertJsonStructure([
      "status", "data", "message"
    ]);
```

```
    }
    public function test_user_cannot_delete_item(){
      $token = $this->user->createToken("api-token")
          ->plainTextToken;
      $this->deleteJson(
        "/api/v1/items/1",
        [],
        ["Authorization" => "Bearer $token"]
      )->assertStatus(403);
    }
    public function test_admin_can_delete_item(){
      // create item
      $item = \App\Models\Item::factory()
          ->create(["category_id" => 1]);
      $token = $this->admin->createToken("api-token")
          ->plainTextToken;
      $this->deleteJson(
        "/api/v1/items/{$item->id}",
        [],
        ["Authorization" => "Bearer $token"]
      )->assertStatus(204);
    }
  }
```

5.2. Siapkan Factories (jika belum):
CategoryFactory & ItemFactory di database/factories

5.3. Jalankan test:

```
php artisan test --filter=ItemApiTest
```

6. Audit Dependency
Di terminal project:

```
composer audit
```

Tinjau daftar vulnerability. Jika ada, perbarui paket:

```
composer update vendor/name
```

Commit perubahan composer.json/composer.lock.

7. Commit & Push

```
git add .
git commit -m "Add tests, logging, rate limiting & dependency
audit"
git push -u origin feature/testing-debug-security
```

8. Pull Request

8.1. Buka Repository di GitHub

- Pergi ke https://github.com/USERNAME/inventory
- Kamu akan melihat notifikasi kuning: "**feature/testing-debug-security had recent pushes**" → klik "**Compare & pull request**"

8.2. Isi Form Pull Request

- Base branch: main (branch tujuan merge)

- Compare branch: feature/auth-sanctum (branch kamu)
- Title: Isi judul PR
- Setelah form diisi, klik tombol "Create pull request"

8.3. Review Kode (Melihat Perubahan)

- Klik tab "Files changed"
- Lihat perubahan kode:
  - Warna hijau = baris kode yang ditambah
  - Warna merah = baris kode yang dihapus

8.4. Merge Pull Request

- Klik tombol "Merge pull request"
- Klik "Confirm merge"